

Emotion Classification Using EEG Data in a Brain-Inspired Spiking Neural Network

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Abstract—As an important application of emotion artificial intelligence, emotion classification provides the basis for the realization of affective brain-computer interface (aBCI). In this study, the NeuCube is used to learn and classify Electroencephalogram (EEG) data from the DEAP dataset. NeuCube is a type of spiking neural network (SNN) framework developed based on the real human brain. It is very suitable for analyzing and processing spatio-temporal data. Based on the 10-fold cross-validation method, we obtain a mean accuracy of 68.91% in the emotional binary valence classification problem. Meanwhile, the EEG data recorded from F3 and F4 electrode channels provide more information compared with Fp1 and Fp2. The results prove that the spiking neural network can be applied to the task of emotion classification effectively.

Index Terms—emotion classification, EEG, spiking neural networks, NeuCube

I. INTRODUCTION

Affective Computing is an interdisciplinary research field involving multiple disciplines such as computer science, psychology, and cognitive science. It aims to research and develop theories and systems that can recognize, interpret, process, and simulate human emotions. Emotional changes are usually produced under the stimulus of the external environment, and will be accompanied by changes in individual representations and psychological responses. Therefore, they can be measured and simulated by scientific methods.

Human emotions are affected by factors such as time, geography, culture, and individual differences. Currently, researchers mainly focus on two types of emotion models: discrete models and dimensional models. The discrete model believes that emotions are composed of a variety of different and discrete basic emotions. The rest of the emotions are a combination of these basic emotions. The true emotions of a person include these basic emotions and other complex emotions. The more famous discrete model is the emotion theory proposed by Ekman [1].

The dimensional model defines emotions in a coordinate system composed of multiple dimensions. Among them, the two-dimensional model of emotions proposed by Russell [2] in 1980 is widely used, which describes all human emotions in terms of valence and arousal by a point in the two-dimensional coordinate system. By using the dimensional method, the emotional response triggered by stimulus can be evaluated and quantified more specifically.

In addition, Zheng et al. [3] showed that there is a certain connection between emotional state, brain neural activity area and Electroencephalogram (EEG) signal frequency band, and the same individual's emotional model based on EEG is stable. The training of model requires a large time cost, because of the high feature dimension of the EEG signal. In order to reduce the feature dimension effectively, Hu et al. [4] used a correlation-based feature selection method in the EEG-based attention recognition task. Zheng et al. [5] proposed to use Group Sparse Canonical Correlation Analysis (GSCCA) for EEG channel selection in the task of EEG-based emotion recognition. Ozerdem et al. [6] used artificial neural networks to conduct research on EEG channel selection in emotion recognition tasks. Clarence et al. [7] used spiking neural networks in emotion classification tasks based on feature fusion of multimodal data and obtained the accuracy achieved by traditional machine learning methods. A greater number of electrodes cannot guarantee a higher accuracy [8].

In this study, through analysis and experiments, we demonstrate a method of using a brain-inspired spiking neural network to perform emotion classification tasks based on EEG data. Specifically, we first encode the preprocessed EEG data (from the DEAP dataset) into a spike train by using the Step Forward (SF) encoding algorithm, and then the hidden spatio-temporal relationships in the spike train are analysed by unsupervised and supervised learning in the NeuCube framework. Finally, test samples undergo supervised learning to achieve binary valence classification. NeuCube is a spiking neural network framework developed based on the real human brain. It can analyze and process spatio-temporal data well.

II. MATERIAL AND METHODS

A. EEG Data Acquisition

In order to verify the feasibility of our study, we use the EEG data from DEAP dataset [9] which can be obtained openly. The DEAP dataset collected the EEG signals (32 channels) and other physiological signals (8 channels) data from 32 healthy participants (50% female). Each participant watched 40 segments of music videos, each with a duration of 60 seconds. The diagram of DEAP dataset can be seen in Fig. 1. The EEG data was first recorded at a sampling rate of 512 Hz under the International standard 10-20 electrode placement system [10], and then down-sampled to 128 Hz. In

order to eliminate the noise caused by eye blinking, all the original physiological signal data had passed a 4-45 Hz band-pass filter. Each array consists of the following two parts:

- *Data*: The data recorded the physiological signals data of each participant during the experiment. Each segment of data was composed of the baseline of the first 3 seconds and the video time of the following 60 seconds. Therefore, each segment of data has a total of 8064 data points.
- *Labels*: The labels recorded the scores (from 1 to 9) rated by participants according to valence, arousal, dominance, and like/dislike corresponding to each video.

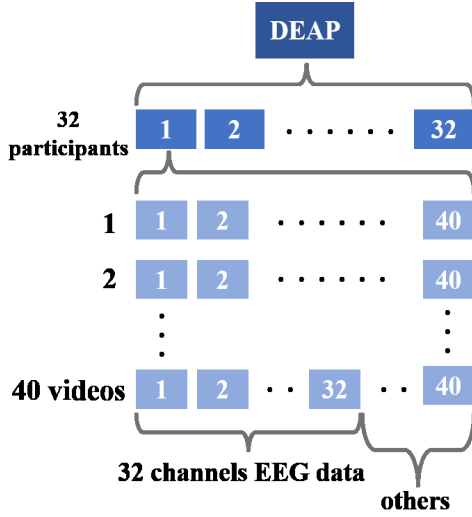


Fig. 1. Diagram of DEAP dataset.

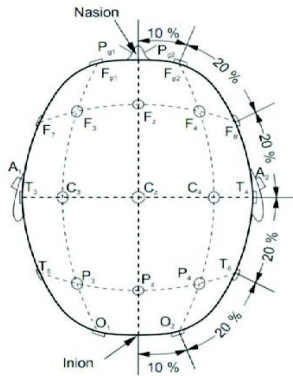


Fig. 2. International standard 10-20 electrode placement system [10].

B. Spiking Neural Networks

As the third-generation artificial neural network model, the Spiking Neural Network (SNN) is more biologically plausible. Compared with the traditional artificial neural network, the spiking neural network can realize the efficient processing of information and the simulation of various neural signals and

arbitrary continuous functions by using the spike firing time of the neuron as the input and output of the network directly.

SNN is a common model for real-world data classification and spatio-temporal pattern recognition. The spatio-temporal data acquired in the real world exhibits spatial characteristics at a certain fixed moment, and at the same time generally exhibits temporal characteristics, such as vision and voice signals. Therefore, the learning and recognition of spatio-temporal data is a challenging task in artificial intelligence, and it is necessary to develop new computational models such as spiking neural network.

C. The NeuCube Architecture

We know that the human brain is extremely efficient in processing spatio-temporal information from the real world. How to analyse the spatio-temporal information efficiently and accurately has always been the direction of scientists' efforts. Traditional machine learning methods are difficult to learn spatio-temporal patterns efficiently based on the structure of the human brain. NeuCube is an extension of eSNN [11]. Its purpose is to convert the spatio-temporal data into spike trains, and then perform unsupervised and supervised learning to realize spatio-temporal pattern recognition. NeuCube not only has been successfully applied to pattern recognition problems based on EEG and fMRI data, but also has further potential in analyzing more variety of spatio-temporal information in the future. A basic NeuCube framework includes the following parts:

- Input data encoding module: This part encodes the input data into a spike sequence, which will be learned in the following part.
- 3D SNN reservoir (3D-SNNr): The encoded spike sequences are presented input into SNNr. All input spike sequences are integrated and converted into class information containing temporal and spatial attributes.
- Classification module: The input verification samples are propagated through SNNr again by using the same supervised learning algorithm as in the learning stage.

D. Encoding Method of Input Data

The filtered EEG data are a train of real-value data. In this paper, the SF encoding algorithm [12] is used to convert EEG data sequences into two types of spike trains: a positive spike sequence (represented as +1) and a negative spike sequence (represented as -1). More details of the SF algorithm can be seen in Algorithm 1. Fig. 3 shows an example of encoding a piece of data (128 data points) in the DEAP database.

E. 3D-SNNr

Different from other architectures of spiking neural networks, NeuCube introduces a real brain structure, which allows the spatio-temporal information in the brain to be analyzed and modeled more effectively. 3D-SNNr is composed of 1471 leaky integrate-and-fire (LIF) neuron models. According to Talairach brain atlas [13], each neuron represents 1cm³ of human brain neuron. At the same time, each neuron is

Algorithm 1 The Step Forward (SF) encoding algorithm**Input:** signal S , baseline B , threshold Thr **Output:** spike train $Spike$

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1:  $L = \text{length}(S)$ 
2: Initialize baseline:  $B(1) = S(1)$ 
3: Initialize spike train:  $Spike \leftarrow 0$ 
4: for  $t = 2 : L$  do
5:   if  $S(t) > B(t-1) + Thr$  then
6:     positive spike :  $Spike(t) \leftarrow 1$ 
7:     update baseline:  $B(t) = B(t-1) + Thr$ 
8:   else if  $S(t) \leq B(t-1) - Thr$  then
9:     negative spike:  $Spike(t) \leftarrow -1$ 
10:    update baseline:  $B(t) = B(t-1) - Thr$ 
11:   others then
12:     no spikes
13:     update baseline:  $B(t) = B(t-1)$ 
14:   end if
15: end for
16: return  $Spike$ 

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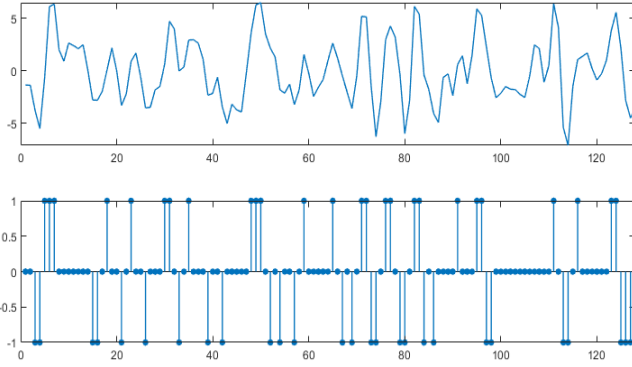


Fig. 3. An example of SF algorithm.

mapped to SNNr, according to the International standard 10-20 electrode placement system. Before learning, the weights between neurons in SNNr will be initialized according to the small-world (SW) rule [11], where 80% of the weights are initialized as positive, and 20% of the weights are initialized as negative. Then the SNNr will be trained in an unsupervised learning mode.

F. Unsupervised Learning

This is the first stage of learning. The purpose is to adjust the connection synaptic weight in the SNNcube so that the spatio-temporal relationship hiding in the input spiking trains can be learned in the entire model. As we know from the Hebbian plasticity rule, if pre-synaptic neurons continue to stimulate post-synaptic neurons, the synaptic weight between them will increase. In this paper, we use the STDP learning rule [14] developed from the Hebbian plasticity rule, that is: if the pre-synaptic neuron i fires before the post-synaptic neuron j , then the connection synaptic weight ω_{ij} between the two neurons

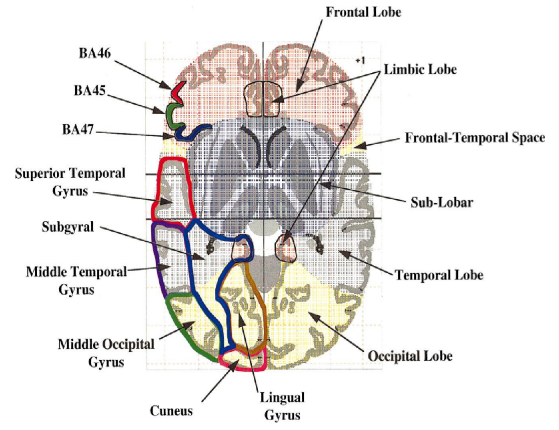


Fig. 4. Talairach brain atlas [13].

will increase, on the contrary, if the pre-synaptic neuron i fires after the post-synaptic neuron j , then the ω_{ij} will decrease. The STDP learning rule in this study is given as [15],

$$\Delta\omega = \text{sgn}(\Delta t) \frac{LR}{\Delta t + 1} \quad (1)$$

where $\Delta\omega$ is the change in the connection weight between post- and pre-synaptic, LR is the STDP Learning Rate (in this paper, $LR=0.01$); $\text{sgn}(\cdot)$ is the function sign, $\text{sgn}(\cdot) = 1$ for positive, while $\text{sgn}(\cdot) = -1$ for negative values; Δt is the difference between post- synaptic and pre-synaptic times and $\Delta t = t_{\text{post}} - t_{\text{pre}}$.

G. Supervised Learning

This is the second stage of learning. After integrating the training sample labels into SNNr, a classifier module is trained through supervised learning. This work uses dynamic evolving spiking neural network (deSNN) [16] as the supervised learning algorithm at this learning phase. The deSNN algorithm combines rank-order (RO) [17] learning algorithm and STDP learning rule. The RO algorithm assumes that the earlier spikes are more important, the first spike carries the most information. After the neuron receives the spikes, it will adjust the weight according to the spikes arrival sequence and time difference. The rate at which the weights are adjusted depends on the setting of parameters values.

The method framework of this article is shown in Fig. 5.

III. EXPERIMENTS AND RESULTS**A. EEG Data Preprocessing**

Due to the complexity of the acquisition method, EEG signals are prone to external interference, often with a lot of noise. Therefore, the preprocessing of the original EEG signal is very important. Effective preprocessing can remove the noise in the original EEG signal, improve the signal quality, and help obtain better classification results. However, more electrodes can not get higher accuracy, only 4 channels (Fp1, Fp2, F3, F4) can also get high classification accuracy [18]. Therefore, we chose four electrode channels, which included Fp1, Fp2, F3, F4. According to the dimensional model of

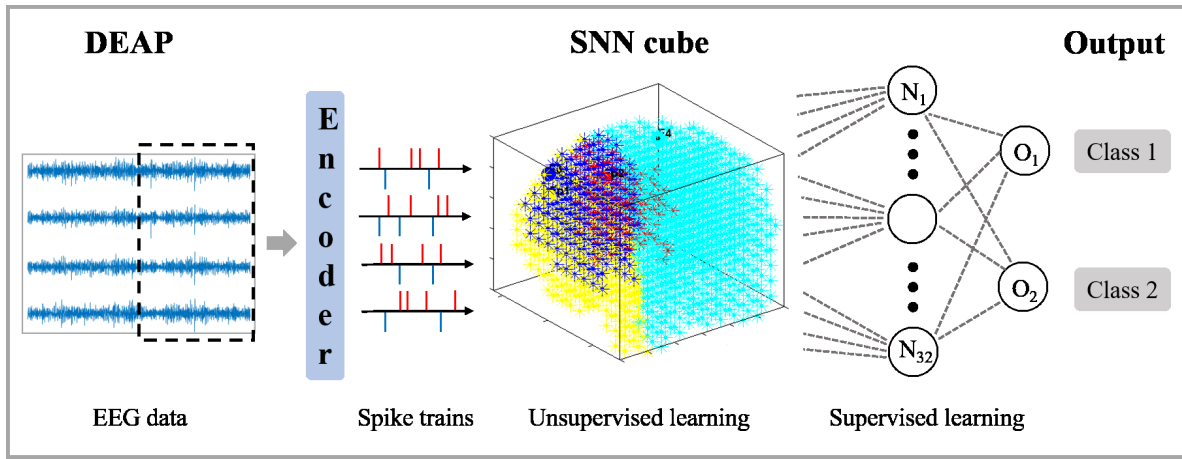


Fig. 5. Diagram of emotion binary valence classification using NeuCube.

emotion classification [2] (as shown in Fig. 6), we marked a valence higher than 6 as positive, and a valence lower than 6 as negative, and removed the baseline in the first 3 seconds. According to [19], the EEG data in the last 30 seconds improves classification accuracy even more, so we only select the EEG data in the last 30 seconds (3840 data points). Finally, we got 1270 samples as our experimental database.

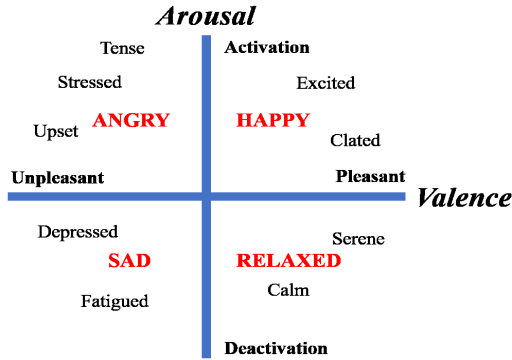


Fig. 6. Dimensional model of emotion classification [4].

B. Classification Method

The input verification samples are propagated through SNNr again by using the same supervised learning algorithm deSNN as in the learning stage. Then the verification samples are classified by using the K-nearest neighbor algorithm. Specifically, if most of the K (in this paper, $K=3$) nearest samples of a sample in the weight space are of a certain class, then this sample is also judged as that class. In order to get more accurate results of classification, the model is evaluated by using the 10-fold cross validation.

C. Optimisation of Results

It is crucial for an emotion classification task to get better classification results by optimizing. In this study, we used

the grid search method. By setting the minimum, maximum, and step length, all target parameters are tested to find the relatively best combination of parameters. After 320 iterations, the relatively better classification accuracy is obtained, and the final parameter combination is:

- Small world radius (γ) : 2
- STDP learning rate (LR) : 0.01
- Threshold of firing: 0.5
- Potential leak rate: 0.002
- Refractory time: 6
- Mod: 0.8
- Drift: 0.05
- K:3

D. Results

After optimization of result, the parameter combination is fixed. There are 32 subjects in all, and each subject has 40 samples. Samples of one subject are presented to NeuCube with fixed parameters. The data of samples will be learned in NeuCube according to the aforementioned process. In the end, we get the accuracy of each subject in this binary valence classification task (see the Fig. 7). The final result is the average of the accuracy of 32 subjects, which is 68.91%, and the maximum classification accuracy is achieved as 82.50%.

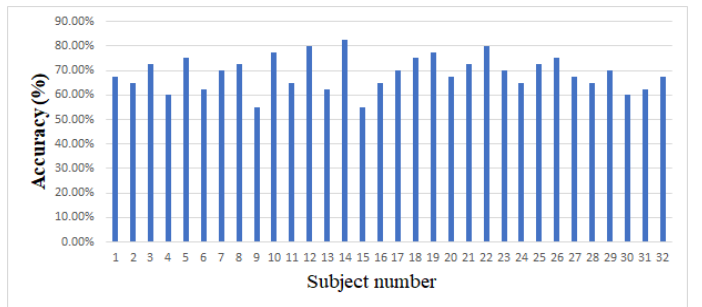


Fig. 7. The classification accuracy of each subject.

It can be seen in Fig. 8 that in this task the input neurons F3 and F4 have the largest clusters, therefore they are the most important features. The Fp1 and Fp2 features are important as well, as their respective clusters are also relatively large.

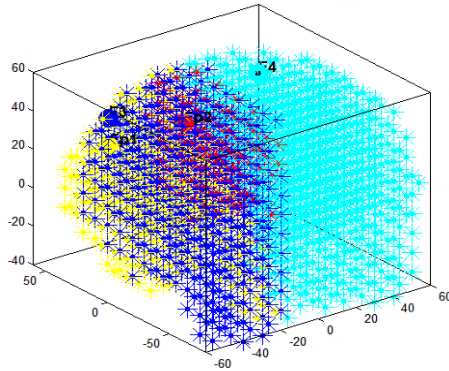


Fig. 8. Neuron clustering for four electrode channels.

To gauge the relative strength of the interaction among neurons more effectively, we get the neuron proportion metric, which is the percentage of neurons in an input neuron cluster out of the total number of neurons in the SNNr. As seen in Fig. 9, the neuron proportion of F3, F4 electrode channels is 31% and 47% respectively compared to 15% and 7% in Fp1 and Fp2 respectively.

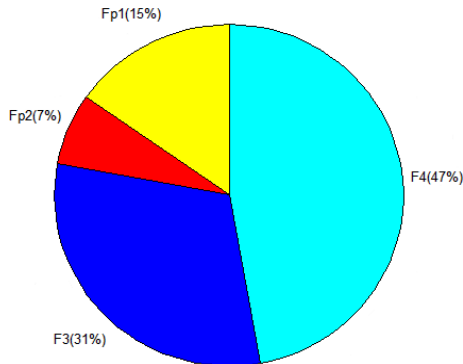


Fig. 9. Neuron proportion for four electrode channels.

IV. CONCLUSION

This paper shows a method based on spiking neural networks for an emotional binary valence classification task by using EEG data. Unlike [14], only EEG data are used in this work. At the same time, in order to balance the performance and computational consumption, only 4 channels of data (the last 30 seconds) are selected. Finally, we obtained an accuracy of 68.91% under the 10-fold cross-validation method. Compared with Fp1 and Fp2, the EEG data recorded from F3 and F4 electrode channels provide more information in this binary valence classification task. The results prove that the spiking neural networks can be applied to the task of emotion classification effectively. Methods based on spiking neural networks are more biologically reasonable compared with

traditional machine learning methods. Despite many efforts, computational consumption is still a limitation of this paper. In addition, we can consider researching the influence of EEG channels and frequency bands to improve accuracy in the future work.

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